Alading Kun

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Education

University of Pennsylvania Aug. 2025 – Dec. 2027 (expected)
Master of Science in Engineering in Scientific Computing

Virginia Polytechnic Institute and State University Sep. 2020 – Dec. 2023
Bachelor of Science in Computational Modeling and Data Analytics
Minors in Computer Science and Mathematics

Publications

M. Saaty, N. Andrus, N. Abdelgawad, J. Chandran, B. Noneman, J. Jackson, **K. Alading**, T. Hassan, D. S. McCrickard, S. Misra, and K. Wernstedt. 2024. "Is Long-distance Hiking an Emotional Roller Coaster?" Evaluating Emotions and Weather Effects on the Appalachian Trail. **CHI EA '24: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems.** Association for Computing Machinery, New York, NY, USA, Article 3, 1–8. https://doi.org/10.1145/3613905.3651024

L. Bilyeu, J. González-Rocha, R. Hanlon, N. Al Amiri, H. Foroutan, **K. Alading**, S. D. Ross, and D. G. Schmale III. 2025. Monitoring Wind and Particle Concentrations Near Freshwater and Marine Harmful Algal Blooms (HABs). **Environmental Science: Advances.** 4, 279–291. Royal Society of Chemistry. https://doi.org/10.1039/D4VA00172A

Research Experience

NASA Marshall Space Flight Center — Huntsville, AL Aug. 2026 – Present
OSTEM Intern | Computational Instrument for Primary Dendrite Spacing Measurement

- Developed a PyTorch U-Net with distance-map prediction and watershed separation to detect dendrite tips in ISS microgravity solidification images; achieved 0.987 F1-score and 1.1-pixel mean center offset relative to reference annotations in five-fold cross-validation.
- Benchmarked against classical image-processing methods (0.828–0.836 F1), evaluated generalization to unseen pulling velocities, and compared temporal U-Net and pretrained SAM-based architectures.
- Quantified detection-error propagation into Voronoi-based primary spacing using 500 Monte Carlo simulations per velocity and method; obtained -0.3% to +0.9% mean spacing bias in idealized arrays, compared with +2.9% to +12.1% for classical baselines.
- Built standalone GPU-accelerated inference software exporting dendrite-tip coordinates, counts, and diagnostic overlays, achieving ∼45 ms/frame end-to-end processing time (∼22 FPS) on a laptop GPU.

NSF IRES at KAIST — Daejeon, South Korea Jun. 2026 – Jul. 2026
LLM-Assisted Pedestrian Route Recommendation

- Designed and formalized a three-layer framework integrating natural-language requirement interpretation, deterministic multi-criteria route optimization, and evidence-grounded recommendation verification.
- Formulated mathematical models for hard-constraint enforcement and weighted subcriteria while preserving user-set priorities across efficiency, safety, and accessibility.
- Designed a bounded LLM architecture that separates interpretation and explanation from deterministic route selection, with capability validation and post-generation verification to support auditable recommendations.

Empathic Computing Laboratory, University of South Australia Apr. 2025 – Sep. 2025
Research Intern (Remote) | EEG Features and Arousal–Valence Modeling

- Conducted cross-dataset EEG analysis with interdisciplinary researchers to study features associated with self-reported arousal and valence for empathic computing applications.
- Applied unified preprocessing and feature extraction, including ICA artifact removal, band-pass filtering, and standardized EEG feature computation.
- Used statistical modeling and machine-learning methods to evaluate EEG feature reliability across multiple

public datasets.

Ross Dynamics Lab, Virginia Tech — Blacksburg, VA Jun. 2023 – May 2025
Forecasting Red Tide-Induced Respiratory Irritation on Florida's Gulf Coast

- Integrated 125K beach observations, 104K *K. brevis* water samples, and meteorological data to develop a daily forecasting dataset covering 40 Southwest Florida beaches.
- Developed an ensemble of LightGBM, PyTorch neural networks, and logistic regression to forecast respiratory irritation severity 1–14 days ahead, achieving 0.79 average precision at the 1-day horizon versus 0.67 for the persistence baseline.
- Designed an early warning system detecting 80–91% of irritation events across severity levels at a 4% false-alarm rate, identifying emerging outbreaks before beach reports.
- Analyzed environmental factors influencing respiratory irritation and validated model performance through historical backtesting, statistical confidence intervals, and comparisons with the published Ross et al. (2022) model.

Virginia Tech — NIST-Sponsored Capstone Sep. 2023 – Dec. 2023
Student Researcher | Nanoparticle Tracking with Artificial Intelligence

- Utilized machine learning to detect and track fluorescent nanoparticles accurately, analyzing their travel distance and characteristics across sequential images.
- Led the design and development of a computational simulation for particle movement, incorporating parameters such as temperature and viscosity.
- Reported approximately 1 nm nanoparticle-radius estimation error in the project evaluation.

Notification Systems Lab, Virginia Tech Sep. 2023 – Dec. 2023
Appalachian Trail Digital Data Analysis

- Conducted emotional analysis of digital data from social media platforms using a RoBERTa-based model, focusing on user-shared experiences of the Appalachian Trail.
- Collaborated with an interdisciplinary team to identify factors influencing emotions through statistical tests, contributing to sustainable trail experiences and effective trail management strategies.

Natural Language Processing Lab, Virginia Tech Feb. 2023 – Jun. 2023
MultiInstruct: Multimodal Instruction-Tuning Data

- Contributed to MultiInstruct, a multimodal instruction-tuning benchmark consisting of 62 diverse multimodal tasks in a unified seq-to-seq format covering 10 broad categories.
- Gathered and analyzed data from 21 open-source datasets, and developed scripts to generate distinct tasks; each task is equipped with 5 expert-written instructions to maximize clarity and precision.

Work Experience

University of Pennsylvania, School of Social Policy & Practice Aug. 2025 – Dec. 2025
Graduate Teaching Assistant | MSSP 6070: Practical Programming for Data Science

- Supported instruction for *MSSP 6070: Practical Programming for Data Science*, teaching graduate students data science, machine learning, and practical programming through hands-on coursework and technical support.
- Guided students in programming for data analysis, modeling, and problem-solving, including preprocessing, model development, evaluation, and interpretation.

Brainy n Bright — Princeton, NJ Jan. 2025 – Aug. 2025
STEM Instructor

- Taught K–12 students programming, 3D modeling, and introductory AI through project-based instruction and personalized mentoring to build confidence and computational thinking.
- Designed and delivered hands-on workshops integrating coding, robotics, creative design, and real-world applications of technology.

Advanced Clinical Trial Supply Ltd — Piscataway, NJ Jun. 2022 – Apr. 2024
Assistant Operations Manager / IT Analyst

- Supported clinical-research Drug Depot Services, including storage, sub-packaging, blinding, labeling, distribution, and recycling; developed SOPs for consistent, documented operations.
- Developed and implemented software to automate drug-distribution and management data processing, supporting efficiency, accuracy, and traceability.
- Conceptualized a warehouse management system, designed a website, and managed drug-storage cloud and office systems.

Lisen Automation — Hamilton, NJ / Los Angeles, CA Jun. 2021 – Aug. 2021
Fitter / Software Testing Assistant

- Assembled a 60-meter package-sorting machine and adjusted its mechanical and electrical components for functionality and reliability.
- Collaborated with a remote software team on software testing, system improvements, and troubleshooting.

Skills

Programming: Python, C/C++, Julia, Java, C#, SQL.
Libraries: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, SciPy, scikit-image, Matplotlib.
Methods: Image segmentation, watershed separation, transfer learning, cross-validation, time-series forecasting, statistical analysis, signal processing, numerical modeling.
Tools: Git, Linux, Docker, CVAT; parallel computing with OpenMP and CUDA.